

BRIAN COHN PH.D.

MACHINE LEARNING ENGINEERING MANAGER, RESEARCH DIRECTOR

[GitHub.com/bc](https://github.com/bc) | [Linkedin.com/in/cohn](https://www.linkedin.com/in/cohn) | [Publications](#)

PROFILE

Hello! I'm a driven research scientist with **a decade of industry experience** revolutionizing healthcare with AI. My expertise spans from designing and running pharmaceutical clinical trials as a PI to coding new AI-based digital instruments/products. As a Director of Research, I've crafted fundable strategies, hired and built expert teams, and led each company I work with towards market leadership in machine learning. I'm excited to leverage this background to enhance your organization's IP portfolio and establish a pioneering position in machine learning research. I'm a unique and motivated scientist, who's made a career out of changing the way we do healthcare—as a machine learning IC, a manager, and a research director.

RELEVANT EXPERIENCE

CEO

Kaspect (Denver)

January **2015** - Present

Independent Machine Learning consultant to multiple companies, offering expertise in IP portfolio strategy and contract work for data science projects.

Key projects:

- Delivered contracted AWS pipeline, FastAPI dashboard, and Django clinician interface as part of a subaward of an SBIR to **Neuromuscular Dynamics LLC**
- In 2015, I identified the SVM/Scala/Spark bug that was keeping **Toyota Motor Sales'** machine learning model from effectively classifying dealership concerns in their C360 project.
- Delivered a 12-feature GBM biomarker model for **Fibronostics** that outperformed the state-of-the-art liver fibrosis model AUC by 0.12 (R)
- Conducted feasibility studies on Mining Social Media Data to influence Patient-Centered Pharmaceutical Product Design for **Eli Lilly** through a Keck Graduate capstone program (Python).
- Invented a spatiotemporal tunneling paper with novel linear programming bootstrap techniques. ([Front. Rehabil. Sci. 2023](#))

Director of Research

ObvioHealth (New York)

January 2021 - July 2023

- Established and led the R&D Department with **7 direct reports:** 2 AI Engineers, 1 data scientist, 1 full-stack developer, 1 game developer, and 2 clinical research assistants. Developed a strong, fun, agile team through 1:1 coaching, mentorship, and hands-on technical guidance. Implemented operational efficiencies and structure in engineering processes from inception that affected the entire startup company.

Notable Products:

- **LLM and RAG**-powered clinical trial writer utilizing proprietary databases and fine-tuned models, streamlining workflows for science writers.
- **AR + ML digital instrument** for characterizing and quantifying diapered stools, featuring a cutting-edge segmentation algorithm. Responsible for drafting and submitting all documentation to the FDA CDER/MDDT. Integrated it into a sponsor's >\$2M pediatric clinical trial with a [pending patent](#).
- **Audio AI suite** for characterizing infant behavior with colic, leveraging Wav2Vec2, whisper, and custom transformer models. Orchestrated a technical integration with LENA SP hardware and software, leading to its adoption in a >\$1M pediatric clinical trial. The code was applied to develop a Cough diarization product, and yielded a paid royalty license.

Doctoral Research Intern

Microsoft Research

June 2019 - August 2019

- Developed Unity C# VR experiments and analyzed the signals in R.
- Co-discovered the VR 'follower effect' → 3 [peer-reviewed papers](#)

CONTACT

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Denver, CO
[Easy-Book Meeting Link](#)

EDUCATION

University of Southern California
•Ph.D. Computer Science, '20
•M.S. Computer Science, '18
Pitzer College
•B.A. Computational Biology, '15

SKILLS

ML (LLMs, Linear Programming, Audio and Video AI), JIRA, Med-Tech & FDA Regulation, Clinical Trial Business Strategy
Python | R | Unity-C# | JS | Scala

NOTABLE AWARDS

NSF Graduate Fellowship
USC Merit Top-Off Fellowship
Grand Prize, MIT Media Lab Reality Virtually Hackathon
Grand Prize, USC - Alexa Prize
Grand Prize, USC Voice Hackathon
Grand Prize, USC Virtual Medicine Competition